

Title

Vascular stiffness and inflammatory response to adjunctive low-dose anti-inflammatory therapy in hypertensive patients: results from the COHERENT trial

Abstract

Background: Inflammation and arterial stiffening are implicated in hypertension pathophysiology. This double-blind, placebo-controlled trial evaluated whether adjunctive low-dose anti-inflammatory therapy reduces vascular dysfunction in hypertensive patients. **Methods:** 150 hypertensive adults (mean age 64.5 years, mean systolic BP 140 mmHg) were randomized 1:1 to receive adjunctive 0.5 mg daily anti-inflammatory agent (n=75) or placebo (n=75) for six months, alongside standard blood pressure medications. Primary endpoint was change in carotid-femoral pulse wave velocity. Secondary endpoints included office and ambulatory blood pressure, echocardiographic parameters, and serum inflammatory and cardiac biomarkers. **Results:** At six months, the treatment group demonstrated numerically greater reduction in pulse wave velocity compared to placebo; however, statistical significance was not achieved (between-group difference -0.34 m/s, 95% CI -0.89 to 0.21). Blood pressure reduction was comparable between groups. Secondary analyses showed modest reductions in high-sensitivity C-reactive protein in the treatment arm. Left ventricular mass and other echocardiographic parameters did not differ significantly. **Conclusions:** In hypertensive patients without established cardiovascular disease, adjunctive low-dose anti-inflammatory therapy did not produce significant improvements in arterial stiffness or other measured cardiovascular parameters over six months.